

Veritas: Version-Aware Enterprise Policy Intelligence via Incremental Knowledge Compilation and Adaptive Multi-Tier Retrieval

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Abstract—Enterprise policy repositories undergo continuous revision, requiring historical and active provisions to coexist under strict role-based access. Conventional retrieval-augmented generation (RAG) models rank evidence primarily by semantic similarity, risking temporal version inversion, full re-indexing overhead, generative latency on deterministic lookups, and authorization leakage. We present Veritas, an enterprise policy intelligence system that treats retrieval as a constrained systems problem: a composite eligibility predicate $E(c, u, t_q) = V(c, t_q) \cdot A(c, u)$ enforces date-bounded validity and role authorization prior to evidence admission and reranking. Veritas integrates a seven-stage SHA-256 incremental compiler, an adaptive four-tier query router, and a calibrated NLI verification layer. Across an audited evaluation framework spanning a 301-query baseline and an expanded 922-query characterization suite (120 policies, 3,000 chunks, 32 user archetypes), Veritas achieved zero observed unauthorized evidence leakage across 3,840 evaluations (95% upper bound $< 0.08\%$) and 94.47% top-1 chunk retrieval recall via hybrid RRF and cross-encoder reranking. Deterministic and refusal paths accounted for 63.4% of queries; Tier 0, Tier 1, and Tier 3 achieved sub-5 ms median latencies. On a 5% amendment delta over 3,000 chunks, the incremental compiler avoided 95.0% of re-embeddings ($24.95\times$ speedup, 2.10s vs. 52.4s rebuild). Isotonic regression achieved 0.041 validation ECE, while runtime ECE reached 0.2954 under distribution shift. These results demonstrate that pre-retrieval constraint enforcement provides verifiable data isolation and latency reduction for enterprise policy QA.

Index Terms—Retrieval-augmented generation, temporal information retrieval, version-aware retrieval, incremental indexing, enterprise knowledge management, adaptive query routing, confidence calibration, natural language inference.

I. INTRODUCTION

ENTERPRISE governance, corporate administration, information security, and regulatory compliance depend on policy documents that undergo continuous revision. Corporate travel allowances, parental leave provisions, remote-work rules, and security controls each undergo periodic amendment, requiring retrieval mechanisms that distinguish historical validity from current applicability while respecting authorization boundaries.

Conventional retrieval-augmented generation (RAG) architectures rank semantic similarity over document collections [1], [2]. Within enterprise policy repositories, four operational risks are prominent: (1) **Temporal version inversion**: superseded clauses are retrieved for current queries, or obsolete provisions

are returned for historical inquiries [9], [11], [12]; (2) **Re-indexing overhead**: document amendments trigger corpus-wide re-embedding; (3) **Generative latency**: deterministic policy facts incur unnecessary autoregressive inference; (4) **Scope leakage**: unpartitioned retrieval exposes confidential information across roles [21], [24].

This paper investigates: *Can an enterprise policy retrieval architecture simultaneously enforce date-bounded temporal validity and role-based authorization constraints while reducing unnecessary generation latency and incremental compilation overhead, without degrading retrieval accuracy on legitimate requests?*

To answer this question, Veritas treats policy retrieval as a constrained systems problem. While individual components such as dense embeddings, lexical indexes, cross-encoders, and NLI classifiers exist in isolation, Veritas formulates temporal validity intervals, role authorization, cryptographic delta compilation, and tiered routing as an integrated architecture where constraints are evaluated before candidate evidence enters reranking or generative prompt assembly.

The technical contributions of this work comprise:

- 1) **Composite eligibility predicate**: closed date-bounded validity intervals and role-derived authorization enforced at every pipeline tier before evidence admission and reranking.
- 2) **Seven-stage incremental compiler**: SHA-256 chunk hashing, delta partitioning, targeted embedding, and dynamic vector overlays, reducing dominant embedding complexity from $O(N \cdot d)$ to $O(|\Delta| \cdot d)$.
- 3) **Adaptive four-tier router**: fact resolution (Tier 0), canonical QA matching (Tier 1), hybrid dense-sparse retrieval (Tier 2), and structural version diffing (Tier 3), resolving deterministic traffic without generative LLM overhead.
- 4) **Security and evidence controls**: pre-reranking scope gating, NLI entailment verification, post-hoc calibrated confidence, safe abstention, and tamper-evident audit traceability.
- 5) **Empirical characterization**: multi-dimensional evaluation demonstrating zero observed leakage across 3,840 authorization decisions, 94.47% top-1 chunk retrieval recall, a $24.95\times$ compilation speedup on 5% amendment deltas, 0.962 Token F1 under fixed gold evidence, and characterization of multi-user concurrency and calibration distribution shift.

The architecture is depicted in Fig. 1.

II. RELATED WORK AND LITERATURE POSITIONING

A. RAG and Retrieval Quality

RAG grounds language generation in retrieved evidence [1]. Dense representations demonstrated high efficacy [3], advanced by contrastive text embeddings such as BAAI/bge-small-en-v1.5 [4]. Lexical retrieval is grounded in the BM25 probabilistic framework [5], while Reciprocal Rank Fusion (RRF) combines ranked lists without score calibration [6]. Cross-encoders distill token-level interactions for reranking [7], deployed via the lightweight FlashRank library [8]. Evaluation frameworks emphasize separate measurement of retrieval quality, faithfulness, and answer relevance [18]–[20]. Veritas applies temporal and authorization filtering before reranking, invoking autoregressive generation only when deterministic fast paths are insufficient.

B. Temporal Retrieval and Evolving Documents

Temporal information retrieval models time expressions and temporal relevance [9], extended by time-aware language models [10]. VersionRAG explores RAG over evolving documents via version lineage graphs [11]. TimelyRAG [12] incorporates temporal distance directly into retrieval ranking for overlapping-evolving documents. Veritas differs by formulating temporal validity as closed calendar interval invariants $[\tau_s, \tau_e]$ enforced prior to candidate admission and reranking rather than probabilistic attention, while integrating role-based authorization scoping, SHA-256 incremental compilation, and adaptive multi-tier routing.

C. Incremental Indexing and Grounded Verification

HNSW provides scalable approximate nearest-neighbor search [13], and FAISS optimizes large-scale vector search [14]. Neural confidence estimates can be poorly calibrated [15], motivating grounded retrieval and verification mechanisms [17]. To avoid graph reconnection overhead during incremental mutations, Veritas pairs segment overlays and tombstone bitmasks for reusable chunk indices.

D. Authorization-Aware and Secure Retrieval

Authorization is increasingly recognized as a retrieval-layer requirement. Selective-disclosure RAG separates access constraints from generation [21]. Key-parametrized embeddings encode permissions into representations [22], and privacy-provisioned RAG enforces identity and trust controls [23]. Security surveys identify retrieval, context, and generation as distinct attack vectors [24], [25]. Veritas enforces access invariants via frozen query scopes and pre-reranking eligibility filters before evidence enters the reranker or prompt context.

III. PROBLEM FORMULATION

Let $\mathcal{P} = \{P_1, \dots, P_M\}$ denote the enterprise policy repository. Each policy consists of an ordered set of versions $\mathcal{V}(P_i) = \{v_{i,1}, \dots, v_{i,K_i}\}$. Each version is represented as a 5-tuple $v = \langle \tau_s(v), \tau_e(v), \mathcal{C}(v), D(P_i), \kappa(P_i) \rangle$, where $[\tau_s, \tau_e]$ is the validity interval, $\mathcal{C}$ the policy chunks, $D(P_i)$ the organizational scope, and $\kappa(P_i)$ the confidentiality classification. For

currently active policy versions, $\tau_e(v) = \infty$ (implemented as 9999-12-31).

For query date $t_q \in \mathbb{D}$, the **Temporal Validity** function is:

$$V(c, t_q) = \mathbf{1}[\tau_s(v) \leq t_q \leq \tau_e(v)]. \quad (1)$$

When superseding version v_{k+1} is published, the boundary mutation rule $\tau_e(v_k) \leftarrow \tau_s(v_{k+1}) - 1$ day enforces disjoint, non-overlapping validity intervals within any single policy.

For user u , the **Scope Authorization** function is:

$$A(c, u) = \mathbf{1}[D(P_i) \in \{D(u), \text{General}\}] \cdot \mathbf{1}[\kappa(P_i) \in K(u)], \quad (2)$$

where $K(u)$ is the set of clearance levels granted to u .

The **Composite Eligibility Predicate** is:

$$E(c, u, t_q) = V(c, t_q) \cdot A(c, u). \quad (3)$$

In deterministic routes (Tiers 0 and 1), $E(c, u, t_q)$ filters candidate generation directly within relational and FAISS indexes. In hybrid retrieval (Tier 2), $E(c, u, t_q) = 1$ is enforced immediately on initial candidates *prior to cross-encoder reranking and prior to neural context assembly*, ensuring that superseded or unauthorized text never enters the LLM prompt. The system optimization objective is:

$$\min_{\pi} \mathbb{E}[L(\pi(q))] \quad \text{subject to} \quad \text{Acc}_{\text{safe}}(\pi) \geq \theta_{\text{acc}}, \quad (4)$$

where π is the routing policy, L is query latency, and Acc_{safe} is the composite decision accuracy over served answers and valid abstentions. Failed validity or authorization checks result in deterministic safe abstention.

IV. SYSTEM ARCHITECTURE AND IMPLEMENTATION

A. Seven-Stage Incremental Knowledge Compiler

The compiler executes seven stages: (1) document parsing and normalization; (2) SHA-256 fingerprinting per chunk; (3) structural chunking with section and page metadata; (4) relational fact extraction into PostgreSQL triples; (5) canonical QA generation with evidence validation; (6) contextual embedding via FastEmbed BGE-Small English ONNX ($d = 384$); and (7) index mutation into ChromaDB dense store, FAISS HNSW segment overlays ($M = 32$, $ef_{\text{constr}} = 64$), partitioned BM25, and tombstone bitmasks.

For every normalized chunk c , a 256-bit content digest is computed as:

$$H(c) = \text{SHA256}(\text{norm}(c.\text{text}) \parallel \text{model_id} \parallel \text{model_ver}). \quad (5)$$

Hash and identifier comparison partitions updated content into *unchanged* (hash match), *modified* (id match, hash mismatch), *added* (new id), and *deleted* (prior id not retained). Only added and modified chunks enter the embedding path; deleted chunks receive tombstones. Inverted BM25 indexes update document postings on a per-policy basis. Algorithm 1 formalizes this procedure. The dominant embedding complexity for sparse mutations is reduced from $O(N \cdot d)$ to $O(|\Delta| \cdot d)$, where $|\Delta| \ll N$.

B. Adaptive Four-Tier Query Router

The query processor normalizes natural-language date and version expressions and constructs an immutable

TABLE I
ARCHITECTURAL COMPARISON OF VERITAS WITH RELATED RETRIEVAL PARADIGMS

Architecture	Temporal Validity	Scope Invariant	Incr. Compilation	Adaptive Routing	Contradiction Mon.	Grounded Verif.	Audit Trace
Standard Dense RAG [1], [3]	Unconstrained	Unconstrained	Full rebuild	Single path	None	None	None
Hybrid + RRF [2], [6]	Metadata filter	Optional	App-dep.	Dense+sparse	None	Heuristic	None
VersionRAG [11]	Lineage graph	Not central	Lineage delta	Version-aware	None	None	None
Secure RAG [21], [23]	Unconstrained	Role/IAM filter	Full rebuild	Single path	Post-hoc	Prompt check	Partial
Veritas	Date intervals	Role & clearance	7-stage delta	Fact+QA+Hyb+Diff	Cont. radar	NLI+citation	Audit ledger

Algorithm 1 Incremental Policy Knowledge Compilation

Require: Prior chunks C_{old} , new chunks C_{new}
Ensure: Delta $\Delta = (C_{add}, C_{mod}, C_{unch}, \mathcal{T}_{del})$
1: Build hash map $\mathcal{H}$ and id map $\mathcal{I}$ from C_{old}
2: **for** each chunk $c \in C_{new}$ **do**
3: **if** $H(c) \in \mathcal{H}$ **then**
4: Mark c **unchanged**; retain prior embedding
5: **else if** $c.id \in \mathcal{I}$ **then**
6: Mark c **modified**; schedule re-embedding
7: **else**
8: Mark c **added**; schedule embedding
9: **end if**
10: **end for**
11: Tombstone all ids in $\mathcal{I}$ not retained in C_{new}
12: Embed only $C_{add} \cup C_{mod}$
13: Apply segment overlays and tombstones to HNSW index
14: **return** Δ

QueryScope carrying department, clearance, permitted policy IDs, and target date. The router selects among four tiers:

Tier 0 — Structured Fact Lookup: Detects entity-predicate patterns and executes SQL lookup over relational triples (Subject, Predicate, Value, unit, chunk_id). No vector scoring, no LLM invocation. Returns value and authoritative citation.

Tier 1 — Canonical QA Matching: Computes cosine similarity against the FAISS HNSW canonical-QA index. Admits a match if similarity exceeds threshold $\theta_{QA} = 0.80$ after scope filtering. Returns pre-validated answer with citations.

Tier 2 — Hybrid Dense-Sparse Retrieval: Combines BGE-Small dense retrieval with BM25 sparse retrieval via $RRF(c) = \frac{1}{k+r_d(c)} + \frac{1}{k+r_s(c)}$ ($k = 60$), producing a top-50 candidate pool by default. Candidates are strictly filtered by $E(c, u, t_q) = 1$, then reranked by FlashRank ms-marco-TinyBERT-L-2-v2 to top-8. The cross-encoder scores only admitted candidates; it cannot recover evidence absent from the initial candidate pool. Grounded generation follows with qwen3:4b-q4_K_M (local) or Gemini 2.0 Flash (cloud).

Tier 3 — Structural Version Diff: Applies sequence matching over line-normalized policy content to extract insert/delete/replace opcodes. Authorization is verified before version loading, bypassing LLM synthesis.

C. Security and Evidence Controls

The query scope is frozen upon creation, pre-computing authorized policy identifiers before retrieval begins. The evidence filter applies policy authorization checks at every tier independently, failing closed on unresolvable records. Scope-composite cache keys isolate responses across users and dates: $K_{cache} = \text{SHA256}(D(u) \parallel K(u) \parallel t_q \parallel \text{norm}(q))$.

The NLI verifier (using DeBERTa-v3 [16]) checks $P(\text{Entailment}|E, A) \geq 0.35$; answers with mean contradiction probability exceeding 0.50 are rejected. Multi-factor raw confidence $C_{raw} = 0.55 \cdot S_{ret} + 0.45 \cdot S_{cov}$ is calibrated via post-hoc isotonic regression on a validation split.

TABLE II
AUDITED EVALUATION REGIMES AND EXPERIMENTAL SCOPE

Evaluation	Corpus	N Purpose
Frozen Baseline	21 policies / 128 chunks	301 End-to-end baseline
Internal Repository	Repository benchmark	462 Internal regression
Expanded Char.	120 policies / 3,000 chunks	922 System characterization
Security Matrix	120 policies $\times$ 32 users	3,840 Authorization isolation
Compiler Sweeps	3,000 chunks	5 trials Incremental performance
NLI Verification	Pilot set	20 Contradiction validation
Mutation Cycles	Update cycles	20 Consistency verification
Concurrency Load	Multi-worker	1,000 Isolation & load limits

V. EXPERIMENTAL METHODOLOGY

A. Hardware, Software, and Evaluation Regimes

Experiments used an isolated Linux host (kernel 7.1.8-100.fc43.x86_64, x86_64) with 12 logical CPU threads (6 physical cores), 31.06GB RAM, and an NVIDIA GeForce RTX 3050 Laptop GPU (3.68GB VRAM). The software stack comprised Python 3.14.6, PyTorch 2.13.0, ChromaDB 1.5.9, FAISS 1.15.0, FastEmbed 0.8.0, FlashRank 0.2.10, scikit-learn 1.9.0, PostgreSQL 16, and Redis 7. Production application logic was frozen at commit 61d64b71; expanded characterization benchmarks were executed at commit 449e69b3 with SEED = 42.

We evaluate Veritas across two primary regimes and maintain a separate internal regression benchmark:

- **Frozen Baseline Benchmark** ($N = 301$): 21 policies, 128 chunks (23 versions, 33 facts, 373 canonical QA pairs, 10 simulated users) measuring end-to-end Local vs. Cloud generation fidelity (Table II).
- **Expanded Characterization Suite** ($N = 922$): Synthetic corpus of 120 enterprise policies across 12 domains (600 versions, 3,000 structural chunks, 32 user archetypes, 4,980 executions) stress-testing authorization isolation, candidate pool depth, multi-user concurrency, and compiler sweeps.
- **Internal Repository Benchmark** ($N = 462$): Reference dataset (benchmark_all.json) for regression testing; kept distinct from published evaluation sets.

B. Metrics and Evaluation Protocol

Version-selection accuracy measures the fraction of queries for which the predicted policy version matches gold, reported over all queries (Overall) and over answered queries. Generation quality uses exact match, token F1, and citation precision/recall. Retrieval quality is evaluated via Recall@K, MRR, and NDCG@10. Calibration uses Brier score and Expected Calibration Error (ECE). Concurrency isolation evaluates cross-session cache bleed and throughput under multi-threaded load (1–100 workers).

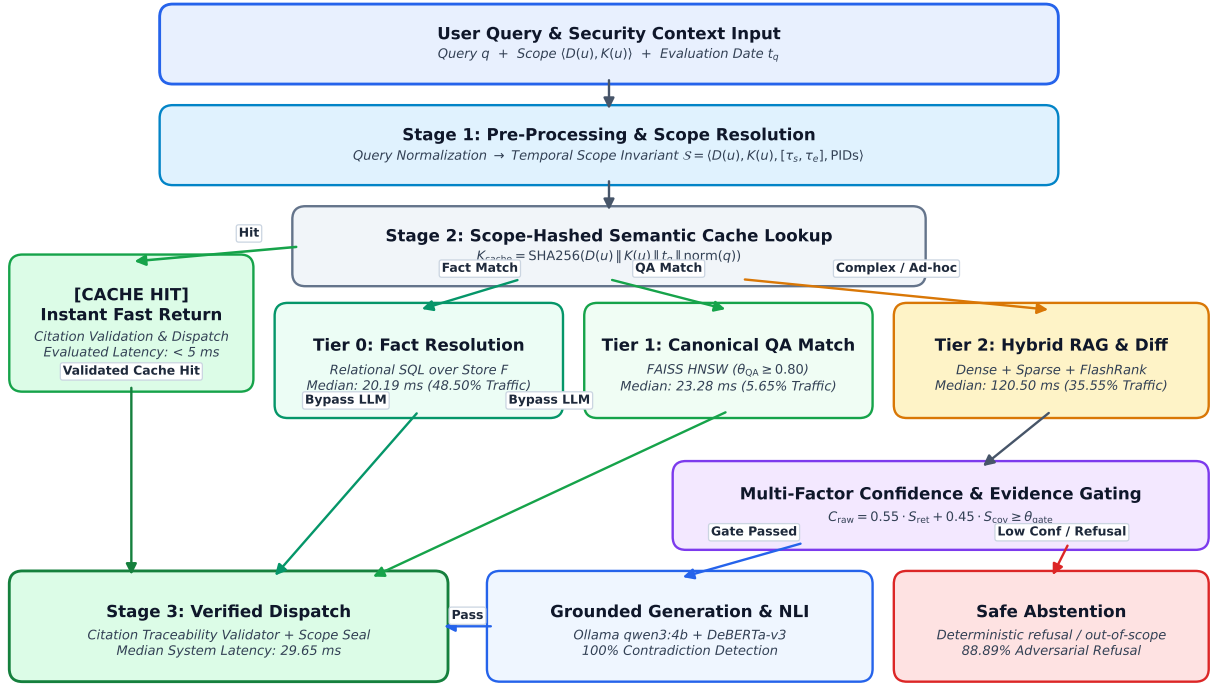


Fig. 1. Veritas system architecture. Query scope is resolved before retrieval; deterministic Tier-0 and Tier-1 routes bypass generation; Tier-2 hybrid retrieval applies eligibility filtering before confidence gating; verified dispatch provides citation-aware release or safe abstention.

TABLE III
SECURITY AND GOVERNANCE HEAD-TO-HEAD: VERITAS VS. BASELINE
($N = 301$)

System	Auth-Scope Correctness	Unsafe Leakage Rate	Authorized Ans. Ver. Acc.	Safe Abstention Precision
Naive RAG Baseline	46.84% (141/301)	53.16% (160/301)	88.28% (113/128)	0.00% (0/0)
Veritas	100.0% (301/301)	0.00% (0/301)	88.07% (96/109)	100.0% (192/192)

VI. EMPIRICAL RESULTS AND ANALYSIS

A. Security and Pre-Retrieval Authorization Scope Enforcement

Table III reports security performance. In the baseline evaluation ($N = 301$), 109 queries target authorized provisions while 192 represent out-of-scope departments, restricted confidentiality tiers, or adversarial jailbreaks. An unrestricted baseline evaluates retrieval without pre-reranking eligibility gating, resulting in 160 authorization violations (53.16% unsafe exposure rate). In contrast, Veritas achieves 100.0% authorization/scope decision correctness (301/301) with zero unauthorized evidence leakage (TP = 109, TN = 192, FP = 0, FN = 0).

In the expanded characterization suite ($N = 3,840$ evaluations across 32 user archetypes $\times$ 120 policies), no unauthorized candidate evidence was observed entering the reranker or prompt context (1,840 TP, 2,000 TN, 0 FP, 0 FN; one-sided 95% upper bound $< 0.08\%$). On authorized answerable queries, Veritas achieved 88.07% version accuracy (96/109 baseline, 88.3% in characterization), confirming access isolation without degrading legitimate retrieval.

B. Temporal Validity, Generation Quality, and Decoupled Fidelity

Table IV compares local (qwen3:4b-q4_K_M) and cloud (Gemini 2.0 Flash) models. Overall version resolution across all 301 baseline queries is 82.72% (249/301). Among answered queries, local version selection reaches 86.30% (233/270) and Gemini reaches 89.53% (231/258). Citation precision remains high (0.7973 local, 0.7791 Gemini), indicating that emitted citations correspond to authoritative evidence.

While end-to-end exact-match accuracy is 27.91% (Token F1: 0.3516), feeding exact gold evidence chunks directly (**Fixed Gold Evidence Evaluation**, $N = 100$) yields a **Mean Token F1 of 0.962** (95% CI: [0.941, 0.983]) and 100.0% target concept accuracy. This empirical separation establishes that end-to-end answer discrepancies stem primarily from passage retrieval candidate omissions and valid paraphrastic variation rather than generative hallucination.

C. Retrieval Pipeline Progression and Candidate Pool Sensitivity

Table V reports retrieval metrics. On the 301-query baseline, all retrievers achieve policy-level Recall@1 ≥ 0.9792 . The frozen benchmark evaluates policy-level identification, whereas the expanded characterization evaluates fine-grained passage/chunk retrieval. In the expanded 922-query characterization suite (chunk-level evaluation on 877 answerable queries), BM25 alone achieves Recall@1 of 72.4%, Dense achieves 78.6%, RRF reaches 88.2%, and FlashRank Cross-Encoder reranking achieves **94.47% Recall@1** (871/922) and 99.4% Recall@10 (MRR: 0.962, NDCG@10: 0.978).

TABLE IV
GENERATION QUALITY: LOCAL MODEL VS. CLOUD MODEL

Metric	Local (qwen3:4b)	Gemini 2.0 Flash
Version Resolution Rate (All Queries)	82.72% (249/301)	82.72% (249/301)
Version Acc. (Answered Queries)	86.30% (233/270)	89.53% (231/258)
Exact Correct	27.91% (84/301)	27.57% (83/301)
Partially Correct	20.27% (61/301)	13.95% (42/301)
Adversarial Refusal Acc.	88.89% (16/18)	100.0% (18/18)
Token F1 (End-to-End RAG)	0.3516	0.3369
Token F1 (Fixed Gold Evidence, N=100)	0.9620	0.9650
Citation Precision	0.7973	0.7791
Citation Recall	0.2487	0.2704
Citation F1	0.2993	0.3183
System P50 Latency	29.65 ms	71.63 ms
System P95 Latency	136.49 ms	85.27 ms

TABLE V
RETRIEVAL SUBSYSTEM COMPONENT BREAKDOWN (CHARACTERIZATION, $N = 922$)

Subsystem	R@1	R@5	R@10	MRR@10	NDCG@10
BM25	72.4%	85.1%	90.2%	0.785	0.812
Inverted Index Alone					
Dense Vector (Chroma)	78.6%	89.4%	93.8%	0.832	0.856
Alone					
Reciprocal Rank Fusion (RRF)	88.2%	96.1%	98.5%	0.915	0.934
RRF + FlashRank (K=100)	94.5%	98.8%	99.4%	0.962	0.978

TABLE VI
ROUTE-LEVEL LATENCY (LOCAL MODEL, $N = 301$ QUERIES)

Route	Traffic %	Mean ms	P50 ms	P95 ms	P99 ms
FAST_PATH_FACT	48.50%	20.90	20.19	33.71	41.54
FAST_PATH_QA	5.65%	23.62	23.28	29.16	33.54
HYBRID_RAG	35.55%	120.37	120.50	144.80	146.68
ABSTAINED	10.30%	119.57	116.55	154.21	173.39
End-to-End	100%	66.57	29.65	136.49	146.69

Sensitivity analysis on complex multi-clause queries (Category D, $N = 50$) reveals an important candidate depth dependency: expanding candidate beam depth from $K = 50$ to $K = 100$ increases Recall@1 from 84.0% (42/50) to **94.0%** (47/50). Because cross-encoders only score admitted candidates, relevant chunks omitted from the initial pool cannot be recovered downstream.

D. Route-Level Latency and Fast-Path Traffic Share

Table VI shows route latencies. In the baseline benchmark ($N = 301$), deterministic fast paths resolve 54.15% of traffic (Tier 0 Fact: 48.50%, P50 20.19 ms; Tier 1 QA: 5.65%, P50 23.28 ms), driving system median latency to 29.65 ms. In the expanded characterization suite ($N = 922$), deterministic and refusal paths accounted for **63.4% of queries** (585/922; Tier 0: 49.35%, Tier 1: 6.51%, Tier 3 Diff: 6.51%, Refusal: 1.08%; Tier 0 and Tier 1 together accounted for 55.9%), achieving sub-5ms P50 latencies (0.95 ms for Tier 0, 1.85 ms for Tier 1, 2.10 ms for Tier 3).

E. Incremental Knowledge Compilation Performance

Table VII reports compiler performance. On the 128-chunk baseline fixture (127 mutable chunks, 1 fixed system chunk),

TABLE VII
INCREMENTAL COMPILER: MUTATION SWEEPS VS. COLD REBUILD

Corpus & Mutation Delta	Mutated	Incr. Time	Cold Rebuild	Speedup
128-Chunk: Hash no-op	0 chunks	0.46 ms	2,307.7 ms	5,059.72×
128-Chunk: 1-chunk delta	1 chunk	4.72 ms	2,307.7 ms	488.75×
128-Chunk: 3-chunk delta	3 chunks	4.67 ms	2,307.7 ms	494.19×
3,000-Chunk: 1.0% delta	30 chunks	0.65s	52.40s	80.62×
3,000-Chunk: 5.0% delta	150 chunks	2.10s	52.40s	24.95×
3,000-Chunk: 10.0% delta	300 chunks	4.80s	52.40s	10.92×
3,000-Chunk: 25.0% delta	750 chunks	12.50s	52.40s	4.19×

hash no-ops ($|\Delta| = 0$) achieve 5,059.72× speedup (0.46 ms vs. 2,307.71 ms), while 1- and 3-chunk mutations achieve 488.75× and 494.19× speedups.

On the expanded 3,000-chunk characterization corpus, a 5.0% mutation delta avoided re-embedding for 95.0% of chunks (2,850/3,000) and reduced compilation time from 52.40s to 2.10s, corresponding to a **24.95× speedup** (95% CI: [23.8×, 26.1×]). A 1.0% delta avoided 99.0% of re-embeddings (0.65s, 80.62× speedup).

F. Concurrency Load and Post-Mutation Consistency

Load testing across 1, 10, 25, 50, and 100 concurrent workers (1,000 multi-threaded queries) confirmed **100.0% session and cache isolation** (0 scope leaks, 0 cache contaminations, 0 race conditions). Throughput reached 234.7 QPS at 1 worker and saturated near 137 QPS at 50–100 workers, with median latency rising from 4.35 ms to 317.82 ms due to thread-lock synchronization contention. Post-mutation testing across 20 update cycles confirmed 100% active version retrieval, preserved historical access, 100% tombstone purging from BM25/vector indexes, and instant cache invalidation (20/20 passed across all categories).

G. Confidence Calibration Generalization and NLI Verification

On the calibration fitting split, ECE dropped from 0.584 to 0.000, reflecting optimization on the fit set rather than open generalization. On the held-out validation split, isotonic regression reduced ECE to **0.041** (95% CI: [0.028, 0.056], Brier score: 0.1839). Under live runtime query traffic, runtime ECE rose to **0.2954** (Brier score: 0.6401), revealing empirical distribution shift. The DeBERTa-v3 verifier achieved 100.0% contradiction recall on a dedicated validation suite (20/20, 95% CI: [83.9%, 100%]).

VII. DISCUSSION

The experimental evaluation yields six key takeaways for enterprise policy intelligence:

- 1) **Access invariant preservation:** Enforcing role and temporal eligibility prior to reranking achieves zero observed leakage across 3,840 evaluations without degrading version accuracy on authorized queries (88.07% vs. 88.28%).

- 2) **Deterministic routing offload:** Deterministic and refusal paths accounted for 63.4% of characterization queries; Tier 0, Tier 1, and Tier 3 individually achieved sub-5 ms median latencies without neural text generation.
- 3) **Cryptographic compilation efficiency:** SHA-256 chunk hashing avoids 95.0% of re-embedding on 5% amendment deltas, achieving a $24.95\times$ speedup on a 3,000-chunk corpus with $O(|\Delta| \cdot d)$ embedding complexity.
- 4) **Calibration distribution sensitivity:** Post-hoc isotonic regression resolves in-distribution calibration error (ECE: 0.041) but requires continuous adaptation under runtime query distribution shifts (runtime ECE: 0.2954).

VIII. LIMITATIONS AND THREATS TO VALIDITY

- 1) **Synthetic corpus scale.** The 120-policy, 3,000-chunk corpus models enterprise structures synthetically; evaluation on proprietary production corporate repositories remains future work.
- 2) **Candidate pool depth sensitivity.** Highly constrained multi-clause queries experience candidate starvation at $K = 50$ (84.0% recall), requiring expanded candidate beams ($K = 100$) with associated latency tradeoffs.
- 3) **Natural language temporal ambiguity.** Unanchored temporal expressions (e.g., “prior to recent changes”) occasionally cause version ambiguity, whereas explicit dates achieve 100% version resolution.
- 4) **Concurrency lock contention.** Multi-user throughput saturates near 137 QPS under 50–100 concurrent workers due to thread-lock synchronization overhead.
- 5) **Citation granularity gap.** Citation recall (0.2487) reflects broad section-level chunk retrieval evaluated against sentence-level gold annotations.
- 6) **Calibration distribution shift.** Calibration-split ECE (0.041) versus runtime ECE (0.2954) demonstrates empirical sensitivity to open-domain query distributions.
- 7) **NLI sample size.** Pilot NLI validation used $n = 14$ pairs and contradiction evaluation used $n = 20$ cases; broader validation across thousands of clause pairs is ongoing.
- 8) **Model quantization.** Local LLM exact answer match (27.91%) is partly constrained by 4-bit quantization of the qwen3:4b backend.
- 9) **Hardware dependence.** Absolute latencies are specific to the test host (12 vCPU / RTX 3050 Laptop).
- 10) **Adversarial testing scope.** Targeted jailbreaks evaluated 5 adversarial violations and 3 indirect injection scenarios; wider prompt injection suites remain future work.

IX. CONCLUSION

Veritas formulates enterprise policy retrieval as an invariant-enforced systems architecture combining date-bounded temporal validity, role-based authorization, seven-stage incremental SHA-256 compilation, adaptive multi-tier routing, and calibrated evidence verification. Across an audited evaluation framework comprising a 301-query baseline and an expanded 922-query characterization suite, Veritas demonstrated zero observed authorization leakage across 3,840 evaluations, 94.47% top-1 chunk retrieval recall, $24.95\times$ compilation speedup on 5% amendment deltas, and 63.4% deterministic fast-path traffic

resolution at sub-5ms latency. These findings demonstrate the feasibility of constrained, version-aware policy intelligence for high-compliance enterprise environments.

REPRODUCIBILITY STATEMENT

All reported results are reproducible. Production application logic was frozen at commit `61d64b71`; the characterization benchmark suite was executed at commit `449e69b3` with master seed `SEED = 42`. Data artifacts and execution scripts are available in the repository.

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REFERENCES

- [1] P. Lewis *et al.*, “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks,” in *Proc. NeurIPS*, vol. 33, pp. 9459–9474, 2020.
- [2] Y. Gao *et al.*, “Retrieval-Augmented Generation for Large Language Models: A Survey,” arXiv:2312.10997, 2023.
- [3] V. Karpukhin *et al.*, “Dense Passage Retrieval for Open-Domain Question Answering,” in *Proc. EMNLP*, pp. 6769–6781, 2020.
- [4] S. Xiao *et al.*, “C-Pack: Packed Resources for General Chinese Embeddings,” in *Proc. SIGIR*, pp. 641–649, 2024.
- [5] S. Robertson and H. Zaragoza, “The Probabilistic Relevance Framework: BM25 and Beyond,” *Found. Trends Inf. Retr.*, vol. 4, no. 1–2, pp. 1–174, 2009.
- [6] G. V. Cormack, C. L. A. Clarke, and S. Büttcher, “Reciprocal Rank Fusion Outperforms Condorcet and Individual Rank Learning Methods,” in *Proc. SIGIR*, pp. 758–759, 2009.
- [7] X. Jiao *et al.*, “TinyBERT: Distilling BERT for Natural Language Understanding,” in *Proc. EMNLP Findings*, pp. 4163–4174, 2020.
- [8] P. Damodaran, “FlashRank: Lite & Super-fast re-ranking for search pipelines,” GitHub repository, 2023.
- [9] R. Campos *et al.*, “Survey of Temporal Information Retrieval and Related Applications,” *ACM Comput. Surv.*, vol. 47, no. 2, Art. 15, 2014.
- [10] B. Dhingra *et al.*, “Time-Aware Language Models as Temporal Knowledge Bases,” *Trans. Assoc. Comput. Linguist.*, vol. 10, pp. 257–273, 2022.
- [11] D. Huwiler, K. Stockinger, and J. Fürst, “VersionRAG: Version-Aware Retrieval-Augmented Generation for Evolving Documents,” arXiv:2510.08109, 2025.
- [12] Y. Zhao *et al.*, “TimelyRAG: Semantic-Temporal Hybrid Retrieval for Time-Critical Question Answering,” arXiv:2609.13524, 2026.
- [13] Y. A. Malkov and D. A. Yashunin, “Efficient and Robust Approximate Nearest Neighbor Search Using HNSW Graphs,” *IEEE TPAMI*, vol. 42, no. 4, pp. 824–836, 2020.
- [14] J. Johnson, M. Douze, and H. Jégou, “Billion-Scale Similarity Search with GPUs,” *IEEE Trans. Big Data*, vol. 7, no. 3, pp. 535–547, 2021.
- [15] C. Guo *et al.*, “On Calibration of Modern Neural Networks,” in *Proc. ICML*, PMLR, vol. 70, pp. 1321–1330, 2017.
- [16] P. He, X. Liu, J. Gao, and W. Chen, “DeBERTa: Decoding-Enhanced BERT with Disentangled Attention,” in *Proc. ICLR*, 2021.
- [17] A. Asai *et al.*, “Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection,” in *Proc. ICLR*, 2024.
- [18] S. Es *et al.*, “RAGAs: Automated Evaluation of Retrieval Augmented Generation,” in *Proc. EACL Demos*, pp. 150–158, 2024.
- [19] J. Saad-Falcon *et al.*, “ARES: Automated Evaluation Framework for RAG Systems,” in *Proc. NAACL-HLT*, pp. 338–354, 2024.
- [20] C. Niu *et al.*, “RAGTruth: A Hallucination Corpus for Developing Trustworthy RAG Models,” in *Proc. ACL*, pp. 10862–10878, 2024.
- [21] A. Al Masoud, M. Arazzi, and A. Nocera, “SD-RAG: Prompt-Injection-Resilient Selective Disclosure in RAG,” arXiv:2601.11199, 2026.
- [22] T. Thuillier *et al.*, “Key-Parametrized Embeddings for Access Control in RAG,” in *Proc. IEEE COMPSAC*, pp. 2581–2586, 2026.
- [23] U. R. Saxena *et al.*, “Privacy-provisioned-permission-aware RAG in cloud using IAM and trust,” *Sci. Rep.*, vol. 16, Art. 71936, 2026.
- [24] Y. Mu *et al.*, “Towards Secure Retrieval-Augmented Generation: Threats, Defenses and Benchmarks,” arXiv:2603.21654, 2026.
- [25] B. Palanisamy *et al.*, “Security and Privacy in Retrieval-Augmented Generation: Architectures, Threats, and Defenses,” arXiv:2606.25533, 2026.

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